More circuits

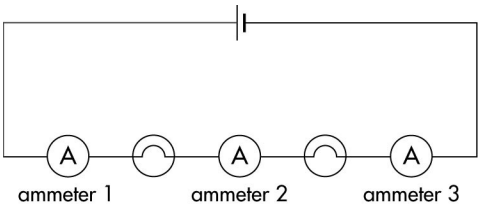
Questions

1 Some students were carrying out an experiment to check whether certain materials were conductors or insulators. Being careful students, they ensured that all components tested were the same size and shape (a fair test). Their results are shown in the following table.

Material	Brightness of bulb	Conductor (C) or insulator (I)?
Copper wire	Very bright	
Steel wire	Not so bright	
Wooden stick	No light	
Lead from a pencil	Very dull	

Complete the last column of the table for the students. What can be said about the structure of the materials tested?

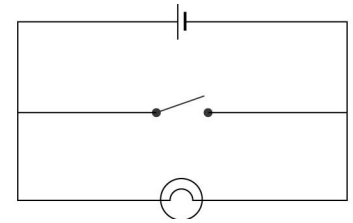
2 In the following circuit, ammeter 1 reads 0.5 A. What will be the reading on:



a ammeter 2? _____

b ammeter 3? _____

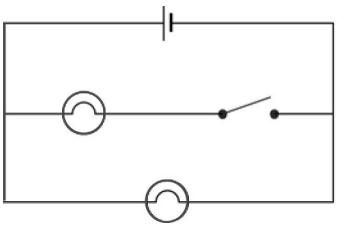
3 For each of the circuits below, say what happens when the switch is closed and try to explain why it happens.



_____

_____

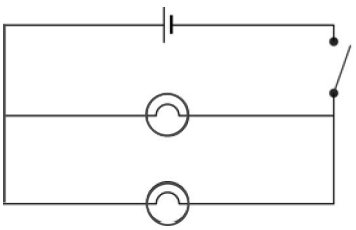
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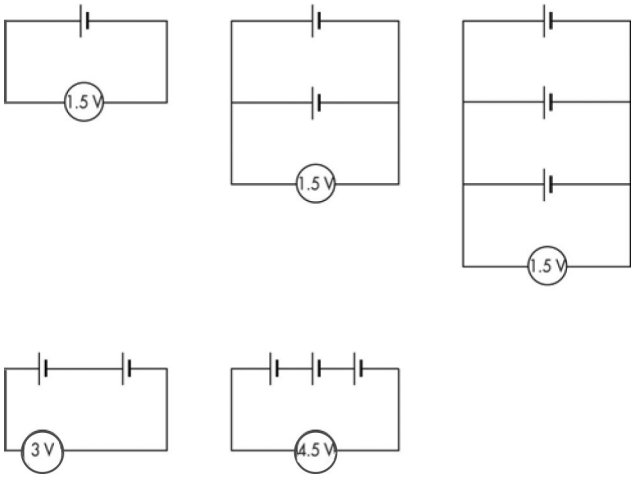
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4 Yvette ran a series of tests on the voltage obtained when torch batteries were connected in various ways. Her results are shown below.



a Yvette has four batteries and wishes to obtain a 6-V output. How should she connect the batteries?

_____

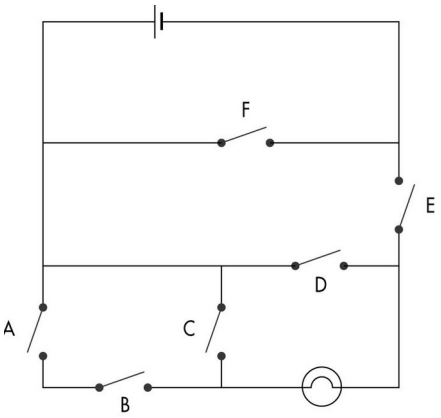
_____

b (A harder one!) If Yvette had to use five batteries and still obtain an output of 6 V, suggest a way of connecting the batteries.

_____

_____

5 Look at the following circuit diagram.



a Will the globe be alight with the circuit as shown? _____

b What is the fewest number of switches that has to be closed to make the light come on? Which switches are they? _____

c If switches A, B and E are closed, the light comes on. What happens if you then switch D on?

_____

_____

d Try to explain your result from part **c**.

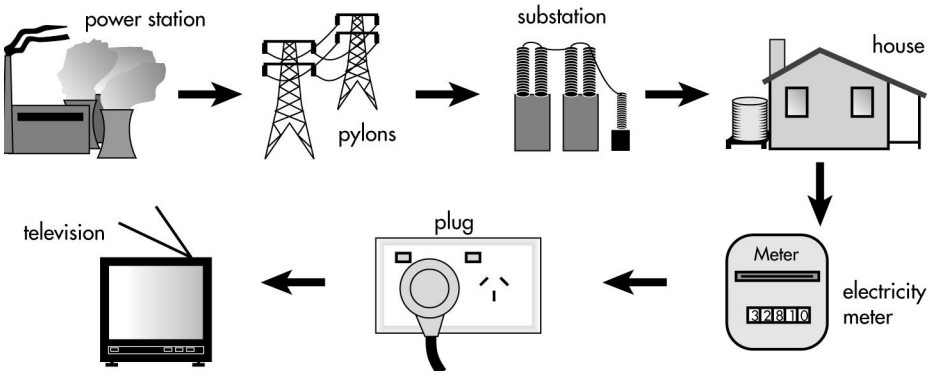
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6 Use words from the diagram below to complete the sentences.



a Electricity is generated in a _____ _____.

b It is carried in big wires or cables. _____ hold the cables above the ground.

c Sometimes the cables are buried in the ground. The electricity always goes to a _____ before it reaches our homes. An _____ _____ outside the house measures how much electricity we use.

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